Paleomagnetism of the Michael Gabbro and Possible Evidence of the Rotation of Makkovik Subprovince

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The Michael gabbro forms a belt of thick, predominantly southerly-dipping sheets which extends 200 km inland from the south-central coast of Labrador. The intrusions lie largely within the Makkovik subprovince, have a radioisotopic age of 1500 m.y., and are reversely magnetized with a paleomagnetic pole position of 163 °E, 10 °N. The paleomagnetic results indicate that most of the rocks internal to this belt have not been deformed since the intrusions acquired a stable remanent magnetization, presumably at the time of cooling 1500 m.y. ago. However, the Michael pole position differs significantly from those of rocks of similar age elsewhere in North America. This suggests rotation of the block containing the Michael intrusions relative to these other rocks. The Aillik dikes which also lie within the Makkovik subprovince have a radioisotopic age of about 1100 m.y. and a pole position consistent with that of rocks of similar age elsewhere in the Canadian Shield. The postulated Michael rotation must then have taken place between 1500 and 1100 m.y. ago.

The rock at two of the most southerly sites sampled for paleomagnetic study proved to be magnetically unstable and the rock at a third such site, though stably magnetized has a magnetization significantly different from the main body of Michael gabbro intrusions. These three sites may indicate a post-consolidation thermal and tectonic event and their position may help define the Grenville Front in this region.

La ceinture de gabbro de Michael est une formation en couches épaisses à pendage sud prédominant qui a une extension de 200 km à l'intérieur des terres à partir de la côte centrale sud du Labrador. Les intrusions s'étendent en grande partie à l'intérieur de la sous-province de Makkovik; elles ont un âge radioisotopique de 1500 m.a. et ont un magnétisme inversé avec un pôle paléomagnétique situé à 163 °E, 10 °N. Les résultats paléomagnétiques indiquent que la plupart des roches contenues dans cette ceinture n'ont pas été déformées depuis l'acquisition par les intrusions d'une aimantation remanente stable, qui coïncide probablement avec leur refroidissement, il y a 1500 m.a. de cela. Cependant, la position du pôle de Michael diffère de façon significative de celles des formations d'Amérique du Nord qui lui sont contemporaines. Ceci suggère une rotation relative du bloc contenant les intrusions de Michael par rapport à ces autres formations. Les dikes d'Aillik, qui s'étendent également à l'intérieur de la sous-province de Makkovik, ont un âge radioisotopique d'environ 1100 m.a. et une position de pôle qui correspond à celles des formations d'âge identique du Bouclier Canadien. La rotation postulée de Michael doit donc prendre place entre 1100 et 1500 m.a.

On a constaté l'instabilité magnétique de la roche, échantillonnée pour étude paléomagnétique, de deux des sites situés les plus au sud; la roche d'un troisième site de position identique aux précédents, bien que magnétiquement stable, a une aimantation sensiblement différente de celle des intrusions de la partie principale du gabbro de Michael. Ces trois sites peuvent fournir l'indice d'une éventuelle post-consolidation thermale et tectonique et leur position peut aider à définir le front de Grenville dans cette région. [Traduit par le journal]

Introduction

Gill (1948, 1949) subdivided the Canadian Shield into a number of fundamental structural units. The boundaries of these units have subsequently been more closely defined and additional units and subunits have been proposed on the basis of new geological and geophysical data (Stockwell 1964). The exact location of the junctions between some provinces has become a popular subject of debate; a debate which is in part semantic since geological and geophysical data are unlikely to provide exactly concordant results. It seems reasonable in terms of geological criteria to position the boundary between older and younger structural provinces in such a way that the younger province contains no significant expanse of rock lacking evidence of having undergone either the pro-

nounced thermal or pronounced deformational events of the orogeny by which the extent of the younger structural province is defined. The presence of radioisotopically older material in a younger province may be used as supporting evidence of a lack of 'younger' thermal alteration. However, the possible loss of argon at rather low temperatures, the possibility of the retention of relict Rb/Sr ages, the possibility of excess argon in potassium-bearing minerals (Stockwell 1963, Wanless *et al.* 1970), and the possibility of widespread deformation without accompanying reheating means that radioisotopic ages, particularly K–Ar ages, should be used with caution in delineating tectonic boundaries. Closely grouped paleomagnetic directions in widespread older rocks may provide evidence of the absence of younger deformation, and in some cases may support radioisotopic data by indicating a lack of later thermal alteration. If, in addition to radioisotopic and paleomagnetic data there is evidence of a structural discontinuity between two orogens, conditions adequate for the delineation of a structural province boundary are realized.

In this context, the exact location of the northeastern limit of the Grenville Front (the boundary between the Nain and Grenville structural provinces) is important because of the possible geological 'fit' of southern Greenland and Labrador. Stockwell (1964) has suggested on geological grounds that this front strikes from lat. 54°35′ N, long. 60°00′ northeasterly to the coast of Labrador. Grasty *et al.* (1969) and Stevenson (1970) have pointed out that rocks yielding pre-Grenvillian K–Ar ages are present south of this line and on the basis of these dates Stevenson (1970) proposed that the line swings eastward parallel and adjacent to the southern border of Benedict Mountains and out to sea in the vicinity of Pottles Bay. Finally, Taylor (1971) suggested that a wedge-shaped area whose southern boundary is shown as a dashed line in Fig. 1 and whose northwest boundary follows Canairiktok River valley be designated the Makkovik subprovince. This was based on the presence of a distinctive northeasterly structural trend within this area. The southern boundary of Makkovick subprovince would constitute the Grenville front in this region.

The difficulties inherent in drawing structural province boundaries on a basis of K–Ar age contours are real because of the lack of universally accepted criteria. In the present instance, some Grenvillian metamorphic ages have been obtained from rocks north of the lines proposed on the basis of K–Ar ages. The question is whether to locate the boundary to the south, until one runs out of old ages, or to keep it to the north of all younger (Grenvillian) metamorphic ages.

A broad pre-Grenvillian easterly-striking belt of basic intrusions that occurs in this region lends itself to a study of the limits of Grenvillian deformation and thermal alteration, and hence to a definition of the Grenville–Makkovik subprovince boundary. The remanent magnetization direction of these rocks is of particular interest as it may provide evidence of the shift of Makkovik subprovince craton relative to Archean Superior craton since the intrusion of these basic rocks.

Geological Setting

Recent reconnaissance mapping by the Geological Survey of Canada (Williams 1970, Stevenson 1970) has outlined the group of basic intrusions which will be referred to informally here as the Michael gabbro for the lake of that name just south of the Benedict Mountains near the south-central coast of Labrador. These intrusions are shown on maps of the region as equidimensional or linear in plan. However, as they probably have the shape of undulatory sheets a probable configuration is suggested on Fig. 1. Their distribution and shape should become more precisely known when aeromagnetic maps become available. The trend of the intrusions suggests that they developed during a period of north–south tension (relative to present geographic co-ordinates).

In the Pottles Bay coastal area the Michael gabbro was described as follows by Kranck (1953, p. 12): "The olivine diabase dykes of Smokey Archipelago, often mentioned in descriptions of the area under 'trap-dykes', usually form steep narrow ridges rising with vertical walls from the light grey gneiss. . . . The mineral composition of a typical medium-grained olivine diabase is: plagioclase, 57.6 per cent; augite 13.8 per cent; olivine, 18 per cent; and ilmenite 9.2 per cent". The prominent sinuous

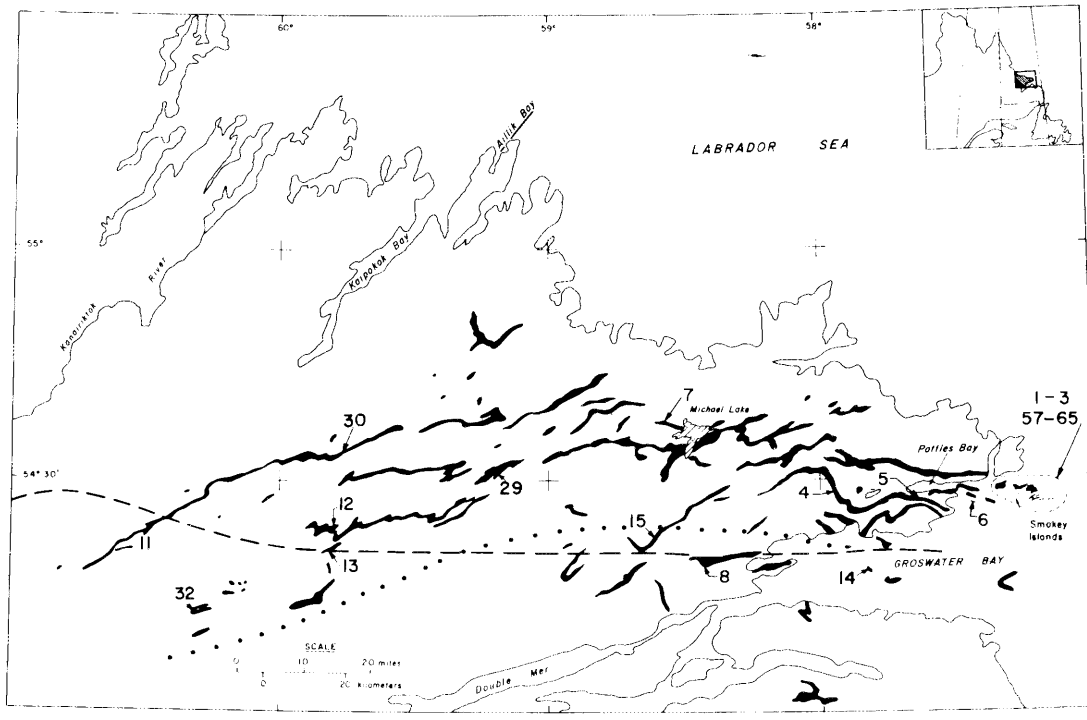


FIG. 1. Distribution of Michael gabbro intrusions modified from Stevenson (1970) and Williams (1970); paleomagnetic sampling localities shown by numbered arrows. Dashed line is the Makkovik subprovince – Grenville province boundary (Taylor 1971) dotted line is this boundary suggested by paleomagnetic data (this study).

ridge south and west of Pottles Bay (see sites 4 and 5, Fig. 1) was described as "massive, coarse-grained, almost black inequigranular poikilitic gabbro composed of plagioclase (70%), olivine (20%), pyroxene (10%)"; intrusions between Tom Luscombe Brook and Double Mer (see site 8, Fig. 1) as "coarse-grained, almost black-weathering, purplish rock composed of plagioclase (50%), perthite and orthoclase (32%), quartz (4%), amphibole (3%), pyroxene (2%), biotite (7%) with traces of sphene, magnetite and anhedral pink garnet", (Stevenson 1970, p. 17).

Paleomagnetic Study

One hundred and thirty-seven oriented drill cores 2.5 cm in diameter were collected at the 25 sites shown on Fig. 1 and listed in Table 1. The cores were oriented in the field by sun compass. Two specimens (2.2 cm in length) were cut from each core and used to yield the mean direction of magnetization of the core. Core mean directions from a site were in turn combined to yield the mean direction at the site and their standard deviation as stated in Table 1. The natural remanent magnetization (n.r.m.) was measured and alternating field (a.f.) demagnetization was carried out at the Geological Survey of Canada using apparatus described by Larochelle and Black (1965), and Larochelle and Christie (1967). The optimum cleaning field for eliminating unwanted secondary components was determined by step-cleaning a single cylinder from each site (from nine sites, one or more additional test specimens were subsequently step-cleaned) in peak fields of 50, 100, 200, 350, 500, and 800 Oe. All but two of the initial 25 test specimens exhibited stable end points which persisted over the same broad range of a.f. demagnetization and the dispersion of magnetization directions of the 23 stable test specimens decreased spectacularly (Fig. 2). This suggested that the primary magnetic component of these 23 sites could be isolated by a.f. cleaning at a common field intensity. The remaining specimens of the

TABLE 1. Directions of remanent magnetization before (N.R.M.) and after (cleaned) magnetic cleaning – Michael gabbro

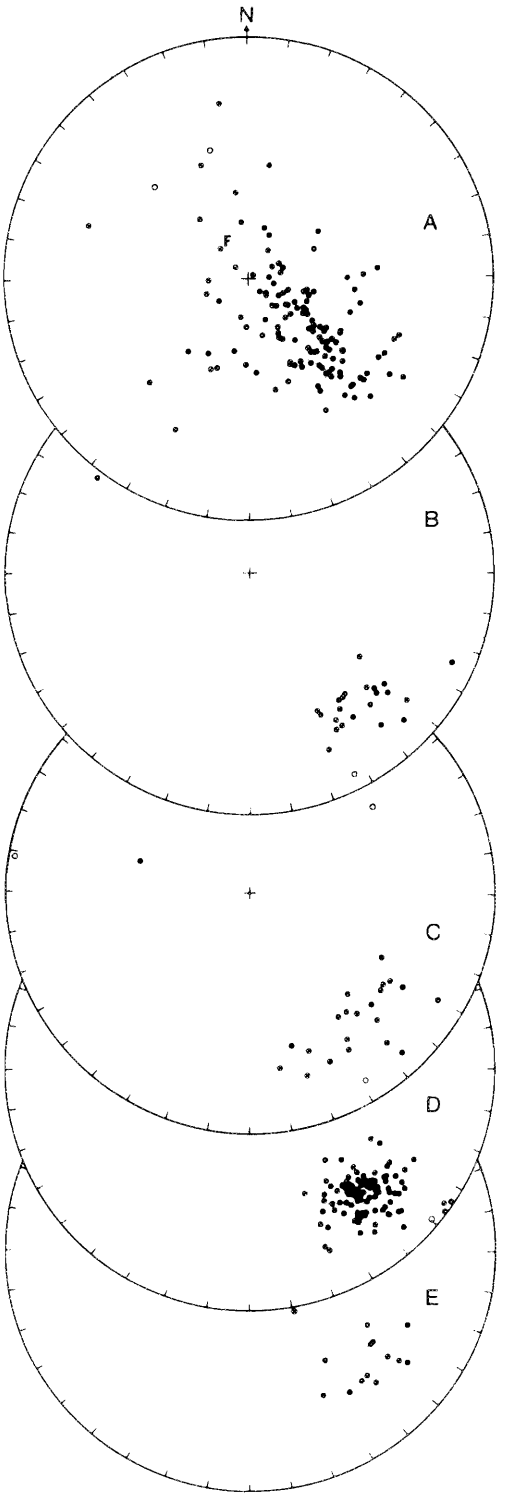
			N.R.M.								
Site	N	n	D	I	$\delta°$	A.F.	D	I	$\delta°$	k	$\delta°_{\bar{R}}$
1	6	5	149	65	13	350	136	35	7.3	124	3.3
2	6	5	121	47	33	350	139	31	7.3	119	3.3
3	7	6	122	66	9.6	350	140	38	5.5	216	2.2
4*	6	2	132	35	26	350	143	24	20	17	14
5	5	5	131	39	9.2	350	134	27	8.2	98	3.7
6	6	6	125	61	12	350	131	29	6.0	180	2.5
7	4	4	154	82	12	350	134	23	3.3	593	1.7
8*	6	1	—	—	—	350	—	—	—	—	—
11	6	6	137	52	4.9	350	138	36	3.7	468	1.5
12	7	7	142	47	6.3	350	146	33	8.4	92	3.2
13	6	4	122	56	19	350	140	32	6.0	184	3.0
14*	6	5	120	39	18	500	128	2.7	7.5	116	3.4
15*	5	0	—	—	—	250	—	—	—	—	—
29	7	4	122	57	19	350	141	26	11	50	5.7
30	6	6	137	38	7.9	350	143	22	8.6	90	3.5
32	7	5	118	71	16	350	131	38	14	33	6.3
57	5	5	33.9	71	23	350	142	35	3.5	528	1.6
58	5	3	252	74	53	350	131	20	4.1	392	2.4
59	5	5	157	76	49	350	140	36	11	58	4.7
60	5	4	136	58	27	350	138	31	7.7	112	3.8
61	5	5	160	51	12	350	143	30	4.5	324	2.0
62	5	4	138	61	29	350	138	30	4.1	382	2.1
63	5	5	125	49	15	350	136	34	3.0	731	1.3
64	5	5	156	65	32	350	139	28	9.3	77	4.1
65	5	5	239	63	46	350	138	38	6.4	163	2.8
All cores	118	104	136	61	27	350	138	32	9.1	79	.9
Accepted sites		21	137	63	20	350	138	31	6.1	161	1.4
Mean pole from 21 accepted site pole positions: 162.5°E, 10.2°N. $k = 226$ $\alpha_{95} = 2°$ $\delta°_{\bar{R}} = 5.4$											
Statistics for 14 cores (whose θ exceed 20°) from 21 acceptable sites											
	14					350	135	35	14	34	
Pole position: 161.9°E, 7.1°N, $k = 34$, $\alpha = 6.2°$, $\delta°_{\bar{R}} = 3.6$.											

*Data from these sites were not included in the group mean for reasons given in the text.
N = number of samples collected at the site, n = number of samples left after eliminating those cores which are inhomogenously magnetized (θ greater than 20°). D = declination, I = inclination, $\delta°$ = angular standard deviation, A.F. = peak alternating field in oersteds, k = Fisher's estimate of precision, $\alpha°_{95}$ = half angle of the cone of confidence at $P = 0.95$, $\delta°_{\bar{R}}$ = standard error of the mean.

23 stable sites were then demagnetized in the optimum field (350 Oe) thus chosen.

The test specimen from site 8 exhibited no end point during step-cleaning so a second specimen from this site was step-cleaned. This specimen also exhibited no end point so the remaining specimens were cleaned at 350 Oe, the optimum field for 23 of the 25 sites. The angle θ between the magnetization directions of two specimens from 5 of the 6 cores collected at this site exceeded 20°. Following Larochelle's (1968) usage, data from this site were not used. The test specimen from site 15 also exhibited no end point so test specimens were run for each of the remaining 4 cores of this site. The optimum field determined by this procedure was 250 Oe, however after cleaning the remaining specimens of this site at this field, the angle θ between magnetization directions of two specimens from every core of this site exceeded 20°. Although the mean magnetization direction of site 4 was in good agreement with the remaining sites, it exhibited a standard error of 14°, so data from this site were also excluded from the final mean. The rock at site 14 is also stably magnetized but the mean direction for this site is significantly different from the other 21 stable sites so data from this site were not used in calculating the mean Michael direction.

In 14 cores from the 21 accepted sites the angle θ exceeded 20° after a.f. cleaning so results from these cores were not used in the final calculations. A single baked contact speci-



men from one of the stably magnetized intrusions gave a magnetic direction after a.f. demagnetization at peak field of 350 Oe, which diverged by 10° from the mean magnetization direction of the intrusions. This provides minor evidence that the gabbros acquired their magnetization at the time of their intrusion.

Analysis of Paleomagnetic Directions

The natural remanent magnetization (n.r.m.) directions of the Michael gabbro sites are generally steeply inclined in a southeasterly direction, (Table 1 and Fig. 2). After a.f. demagnetization in a peak field of 350 Oe within site dispersion generally decreased spectacularly in the 23 stably magnetized sites and site means of the 21 accepted sites were grouped in a southeasterly direction at moderate inclination.[1] The directions of magnetization of test specimens during step demagnetization indicated that most of the change in inclination was accomplished by cleaning at 50 Oe; thereafter cleaning in fields from 100 to 350 Oe effected a steady small decrease in inclination. Demagnetization at 500 Oe did not affect the mean direction of magnetization but dispersion increased slightly and this became more pronounced after cleaning at 800 Oe (Fig. 2). The demagnetization curves (Fig. 3) of test specimens from all but sites 3, 4, 5, 11, 12, and 60 indicate that a large soft component is largely removed by a.f. demagnetization at 100 Oe isolating a stable

[1]Since this paper was submitted, paleomagnetic data were published on 16 cores from dikes near Indian Harbor (Smokey Islands of Fig. 1) by Murthy and Deutch (1972). These samples were undoubtedly collected from the same rocks referred to here as Michael gabbro, however the magnetization direction reported after a.f. treatment (242°, +22°) is almost 90° clockwise from the direction reported in the present paper.

FIG. 2. Equal area projection of Michael gabbro magnetization directions. A. natural remanent directions all specimens, B. test specimen directions after a.f. cleaning in 350 Oe peak field, C. test specimen directions after a.f. cleaning in 800 Oe peak field, D. directions of cores from 21 stably magnetized sites after a.f. cleaning in 350 Oe peak field (excludes 14 cores whose θ values exceed 20° after cleaning in 350 Oe peak field). E. directions of the 14 cores (from stably magnetized sites) whose θ values after cleaning at 350 Oe peak field, exceed 20°.

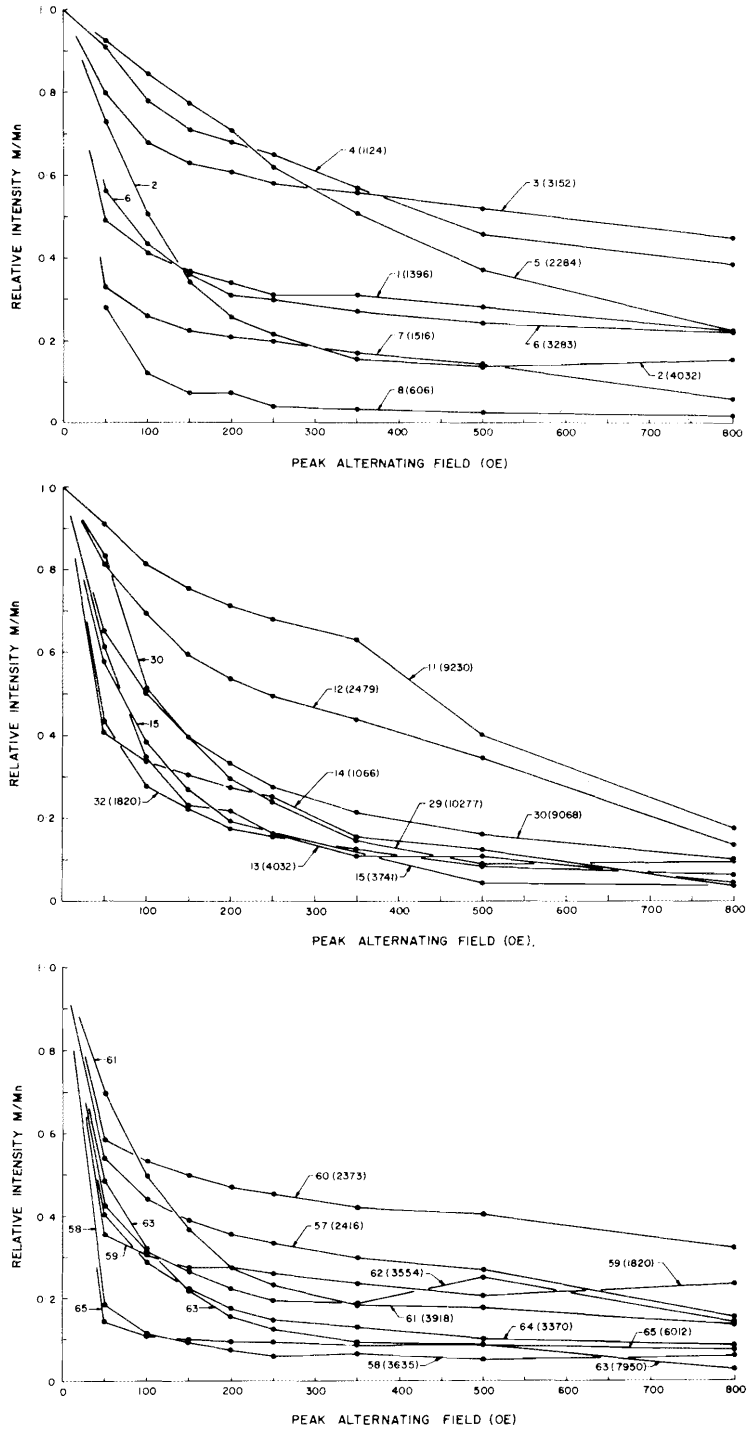


FIG. 3. Alternating field demagnetization curves of one specimen from each Michael gabbro site. The numbers in brackets beside the curves are the n.r.m. intensities in e.m.u. $\times 10^{-6}$ per cm^3 for the specimens.

component. This gradual decline in magnetic intensity over a wide range of a.f. demagnetization and the excellent end points of the test specimens suggests successful isolation of an unstable isothermal component from the harder thermo-remanent component. The concordance of magnetization direction exhibited by these intrusions which occur over an area of 9000 km^2 suggests that the hard component was developed during the initial cooling of the intrusions. The shift of the magnetization direction directly away from the recent earth's field direction suggests that the vector erased during a.f. demagnetization was a soft component related to this field.

The specimens from the 6 most stable sites (3, 4, 5, 11, 12, 60) showed the same stable end points as the test specimens from the other 17 stable sites but did not exhibit a rapid decrease in magnetic intensity in cleaning fields of 50 and 100 Oe. The test specimens from these sites show virtually a straight line decrease in magnetic intensity and retain almost half their initial intensity after a.f. cleaning at 500 Oe (Fig. 3). Of these 6 sites, sites 4 and 5 are from the same intrusion, which forms the continuous gabbro ridge south of Pottles Bay.

It is noteworthy that 3 of the 4 sites excluded from the final calculations of paleomagnetic data lie close to the dashed line that Taylor (1971) has proposed as the southern limit of his Makkovik subprovince and presumably the northern limit of major Grenvillian thermal and tectonic events in this region. The rocks at sites 8 and 15 are magnetically unstable and although the rock at site 14 is stably magnetized and internally consistent the site mean direction differs enough from that of the remaining stable sites, to suggest that this site has been disturbed either thermally or mechanically relative to the major belt of Michael intrusions.

The Michael gabbro has a mean pole (unit weight to each site pole) at 162.5 °E, 10.2 °N, $k = 226$, and $\alpha_{95} = 2°$, $N = 21$. The site poles are rather closely grouped, however, they represent a number of intrusions that by comparison with young, well-dated, dolerite–diabase sequences would be expected to represent intrusion spanning a significant period of time (10^5 to 10^7 yr). Furthermore, some of the intrusions are thick, coarse grained (for example localities 4, 5, 7, and some of those in the Smokey Islands area) and must have cooled over a period of thousands of years so that mean site data from these individual intrusions partly averages the paleosecular variation. For these reasons we feel that the 21 sites provide a good paleomagnetic pole position at the time of Michael gabbro intrusion, and the contemporary paleosecular variation was probably small.

It has been the practice of this laboratory to discard data from samples if θ, the angle between the magnetization directions of two specimens cut from the samples, exceeds 20°. This is parallel to discarding measurements obtained from a site, if the standard deviation of their mean exceeds 10°, except that the criterion of dispersion is applied at the sample, rather than at site level. This practice has been followed in the present study as within-sample dispersion indicates local inhomogeneous magnetization which may in some cases be due to disturbed magnetization of part or all of the sample. We thought it of interest to determine whether the inhomogeneity of the 14 rejected Michael gabbro cores from 21 (accepted) sites was due to random or non-random factors. The mean pole position derived from the directions of these 14 cores (after a.f. cleaning at 350 Oe) turned out to be 162 °E, 7 °N. This appears to indicate that the within-sample inhomogeneity of the Michael gabbro is due to localized random factors and the exclusion of data from these 14 cores has in fact no significant effect on the mean pole position reported above (see Fig. 2).

Age of the Michael Gabbro

The radioisotopic age of the Michael gabbro has been determined by both the K–Ar and Rb–Sr method. The ages obtained by the K–Ar method from rocks that are Michael gabbro (G.S.C. determinations) and from rocks that are probably Michael (Grasty *et al.* 1969) are listed in Table 2. The 2080 ± 42 m.y. age obtained by Grasty *et al.* (1969) from an olivine gabbro dike near Indian Harbor probably represents a Michael gabbro intrusion, however, the authors considered the age to be anomalously great, probably due to excess argon. The other whole-rock, K–Ar age which they obtained on olivine gabbro at this area is 974 ± 13 m.y. This sample was probably also

TABLE 2. K–Ar Whole-rock ages

Sample No.	Location		K%	$\frac{40\text{Ar rad}}{40\text{Ar tot}}$ %	Age	Reference
			Michael gabbro			
1	54°44′ N	57°51′ W	1.13	97.7	974 ± 13	Grasty *et al.* 1969
7	54°27′ N	57°13′ W	1.07	95.9	2080 ± 42	Grasty *et al.* 1969
FA–1–69	54°27′ N	57°17′W	1.11	98.0	1315 ± 50	G.S.C.
FA–5–67	54°24′ W	57°12′ W	1.04	95.0	1340 ± 120	G.S.C.
			Aillik diabase			
FA–2–69	55°12′ N	59°13′ W	4.45	98.0	685 ± 32	G.S.C.
FA–3–69	55°12′ N	59°13′ W	2.42	98.0	935 ± 64	G.S.C.
FA–5–69	55°12′ N	59°12′ W	2.13	97.0	871 ± 64	G.S.C.
8	55°13′ N	59°10′ W	2.08	98.8	995 ± 13	Ghandhi *et al.* 1969
10	55°15′ N	59°21½′ W	1.78	89.4	956 ± 16	Gandhi *et al.* 1969

obtained from a Michael intrusion. The two G.S.C. whole-rock K–Ar ages are concordant and indicate an age between 1300 and 1400 m.y. Previous experience with whole-rock, K–Ar dating of fresh basic intrusions has indicated that argon loss results in ages at least 5 to 10% lower than the more reliable Rb/Sr isochron method. A K–Ar age on Michael gabbro of between 1300 and 1400 m.y. would be concordant with a Rb/Sr whole-rock isochron age approaching 1500 m.y.

A whole-rock Rb/Sr isochron age of 1488 ± 107 m.y. using $\lambda = 1.39 \times 10^{-11}\text{y}^{-1}$ (personal communication R. K. Wanless) was obtained by using material from the chilled margin, coarse center, and three granophyric veinlets from several Michael intrusions of the Smokey Islands archipelago.

The veinlets are restricted to the intrusions and are clearly late magmatic. The reasonable concordance between the K–Ar ages and the Rb/Sr whole-rock age is supporting evidence that the Rb/Sr isochron provides a good estimate of the age of intrusion. It is unlikely that a mild metamorphic event (higher grade metamorphism of the stably magnetized sites is ruled out by a lack of evidence indicating mineralogic changes) would have set the K–Ar and Rb–Sr clocks to give such good agreement between the two systems. The evidence of a single magnetization direction from this broad area of stably magnetized gabbro is also evidence of a lack of secondary thermal effects.

Aillik Dikes

The magnetizations of rocks, both older and younger than the Michael intrusions, from within the Makkovik subprovince are of particular interest. Results are available from one such set of intrusions namely the Aillik dikes, from the Aillik Bay area. These intrusions are much less extensive than are the Michael intrusions. They are those porphyritic dikes described by Kranck (1953, p. 19) as "Coarsely porphyritic with euhedral phenocrysts of plagioclase up to 4 inches long". Paleomagnetic laboratory procedures followed were those used in the study of the Michael samples and paleomagnetic data are summarized in Table 3 (34 samples from 6 dikes). Potassium–argon ages on these dikes are listed in Table 2. These data suggest that the Aillik dikes are in the 1000 to 1100 m.y. range assuming that the highest age (995 m.y.) most closely approaches the age of intrusion and that even this rock has undergone some argon loss. The dikes are normally magnetized and their magnetization yielded a pole position of 136 °W, 27 °N (Table 3) for the period of their original cooling.

Discussion

Concordance of magnetization directions within a belt 200 km long and 50 km wide indicates that the Michael gabbros within this belt, and hence the Archean and Proterozoic country rocks within this belt have undergone neither significant internal tectonic deformation nor thermal alteration since the gabbros were formed, and presumably magnetized 1500 m.y. ago. However, the mean pole position of the stably magnetized Michael rocks is some 30° away from the concentration of poles for rocks

TABLE 3. Directions of remanent magnetization before (N.R.M) and after (cleaned) magnetic cleaning – Aillik dikes

	N.R.M.										
Site	N	n	D	I	δ°	A.F.	D	I	δ°	k	δ_R°
9	6	6	278	45	15	350	278	45	16	26.8	6.4
45	6	6	284	38	6.8	350	284	36	7.4	121	3.0
46	6	5	272	64	8.6	350	272	64	8.6	88.7	3.8
48	6	6	274	41	5.6	350	274	41	5.5	217	2.2
50	6	6	281	62	8.1	350	277	52	3.7	479	1.5
56	4	3	280	65	9.3	350	280	56	2.7	873	1.6
All cores		32	276	57	13	350	278	48	12	42	2.2
All sites		6	276	58	11	350	278	50	10	60	4.3

Mean of 6 site pole positions: 136 °W, 27 °N, $k = 64$, $\alpha = 7.1°$, $\delta_R^\circ = 10$.

NOTE: Symbols used are the same as those in Table 1.

of similar age elsewhere in North America as illustrated in Fig. 4 from Robertson and Fahrig (1971). These data suggest that the block containing Michael gabbro underwent movement relative to other paleomagnetic sites of similar age (and hence to much of the Precambrian of North America) subsequent to the acquisition of a stable magnetization about 1500 m.y. ago.

The pole position of the Aillik dikes, which also lie within the Makkovik subprovince as defined by Taylor (1971) have an interesting bearing on this possible rotation. The pole for these rocks is reasonably concordant with those of similar age elsewhere in the shield (see Fig. 4), for example the Abitibi dikes (Larochelle 1966). The Aillik dikes are normally magnetized as are the Abitibi dikes, whereas the Michael gabbro is reversely magnetized, as the rocks of apparently similar age elsewhere in North America. The major part of the postulated rotation of Makkovik subprovince must then have been accomplished between the time of intrusion of the Michael gabbro, about 1500 m.y. ago, and the intrusion of Aillik diabase about 1100 m.y. ago.

Gabbro along the southern fringe of the stably magnetized Michael rocks has been correlated with them on the basis of contiguity, chemistry, and similar mineralogy but locally shows evidence of having been thermally altered (c.f. Stevenson's 1970 report of the presence of garnet, and evidence cited above indicating unstable magnetization of sites 15

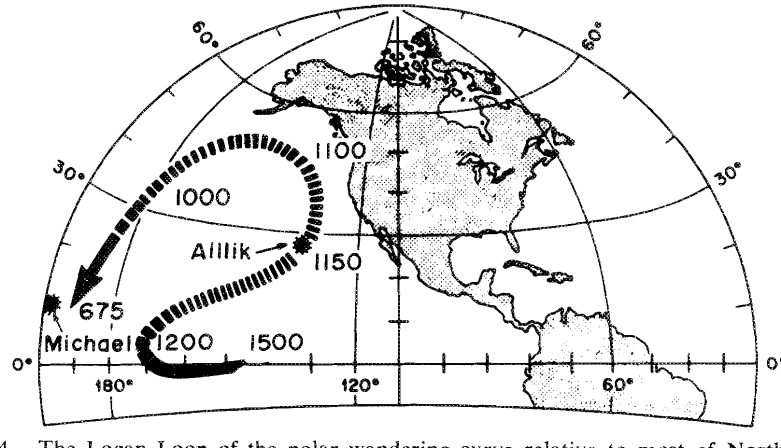


FIG. 4. The Logan Loop of the polar wandering curve relative to most of North America in Hadrynian and Helikian time from Robertson and Fahrig (1971) with the added pole positions of the 1500 m.y. Michael gabbro and 1000–1100 m.y. Aillik dikes. Numbers along the curve indicate the approximate ages of the pole positions used by Robertson and Fahrig in the construction of this curve.

and 8). The location of these altered rocks suggests the possibility of thermal alteration related to the Grenvillian orogeny. Furthermore, the rock at site 14, though stably magnetized has a magnetization direction significantly different from the other 21 stably magnetized sites. This suggests tectonic rotation of this site relative to the other Michael sites. This evidence of thermal and tectonic effects along the southern edge of the Makkovik subprovince may indicate the limits of post 1500 m.y. old orogeny in this area. This limit is indicated by a dotted line in Fig. 1. It differs only slightly from the limit of Grenvillian deformation (dashed line) as suggested by Taylor (1971).

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